Theorem Summary: Exact Repair Obstructions for a Sensitivity–14 Kushilevitz Composition

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Setup and historical base. Let

$$K(y) = \sum_i y_i - \sum_{i<j} y_i y_j + \sum_{T\in\mathcal{D}} \prod_{i\in T} y_i,$$

$$\mathcal{D} = \{013, 014, 024, 025, 035, 345, 145, 125, 123, 234\}.$$

The set $\mathcal{D}$ is the $2-(6,3,2)$ design, and K is a relabeling of the degree-three, sensitivity-six function of Kushilevitz recorded by Nisan and Wigderson. For four disjoint raw triples X_i, put

$$L_i = \sum_{a\in X_i} x_a, \qquad E_i = \sum_{\{a,b\}\subset X_i} x_a x_b, \qquad g_i = \mathrm{NAE}_3(X_i) = L_i - E_i.$$

Near-solution (degree 6, sensitivity 14). The Boolean function

$$P = K(g_0, g_1, g_2, g_3, x_{12}, x_{13})$$

satisfies $P(0) = 0$ and $P(e_j) = 1$ for $0 \le j < 14$. Thus $s(P, 0) = 14$. Its complete highest layer is

$$P_6 = -E_1E_3(E_0 + E_2).$$

It has exactly $2 \cdot 3^3 = 54$ distinct degree-six monomials, all with coefficient -1, and none of higher degree.

Two noncancellation principles. First uniquely multilinearize a nonzero meta-polynomial Q. Under substitution of disjoint NAE blocks and disjoint coordinate ports, distinct meta monomials have distinct raw top signatures. Giving NAE variables weight 2 and ports weight 1,

$$\deg Q(g, z) = \max_{c_{I,J}\neq 0} \big(2|I| + |J|\big).$$

Disjointness and multilinearization are essential.

For the separate Walsh lemma, in spin variables let

$$\chi_i = s_{i0}s_{i1}s_{i2}, \quad u_i = \frac{s_{i0}s_{i1} + s_{i0}s_{i2} + s_{i1}s_{i2}}{2}, \quad r_i = 1 - 2g_i = u_i - \frac{1}{2}.$$

If $q = \sum_{U,B} d_{U,B}\, u_U t_B$, multiplication by $\chi_0\chi_1\chi_2\chi_3$ sends each nonzero sector (U, B) to disjoint Walsh characters of degree

$$12 - 2|U| + |B|.$$

This block-signature statement is not an application of the weighted lemma.

Central selector obstruction. Define

$$A = K(g_0, g_1, g_2, x_9, x_{10}, x_{11}).$$

Then A is Boolean, has degree 5, sensitivity 12, and

$$A_5 = E_0E_1(x_9 + x_{10}) + E_0E_2(x_{10} + x_{11}) + E_1E_2(x_{11} + x_9).$$

There is no Boolean $B : \{0,1\}^{12} \to \{0,1\}$ with $B(0) = 1$ and $\deg(B - A) \le 4$. Averaging B over block permutations and whole-block complements gives a $[0,1]$-valued function of the NAE outputs and ports. Its fixed-port heavy slices are quadratic. The sums S_z of their three pair coefficients satisfy

$$S_{111} = S_{000} + 6, \quad S_{000} \ge -3 + 2C(0), \quad S_{111} \le 3, \quad C(0) \ge \tfrac{1}{8},$$

a contradiction. Here B may depend arbitrarily on all twelve raw variables; the theorem is scoped to the fixed base A.

Gate-factor and fixed-routing obstructions. Let $T : \{0,1\}^6 \to \{0,1\}$ and $F = T(g_0, g_1, g_2, g_3, x_{12}, x_{13})$. If $F(0) = 0$ and $F(e_j) = 1$ for every $0 \le j < 14$, then $\deg F > 5$: setting the ports to zero would leave a Boolean quadratic with four sensitive meta-bits, whereas every Boolean quadratic has sensitivity at most 3. Moreover, let each h_i be a Boolean function of degree at most two with $h_i(0) = 0$, sensitive at its own origin in all three coordinates of a designated pairwise-disjoint triple. Routing h_i to input i of K, with two distinct coordinate ports disjoint from all triples routed to inputs 4, 5, cannot evade the degree-six layer.

Full-parity-twist obstruction. For every Boolean meta-function T,

$$T(g_0, g_1, g_2, g_3, x_{12}, x_{13}) \oplus p_0 \oplus p_1 \oplus p_2 \oplus p_3$$

has degree > 5, where p_i is raw XOR parity on block i. The Walsh lemma leaves only a putative output spin $q_T = a(t_{12}, t_{13}) \prod_i (r_i + 1/2)$. If $a = 0$, Booleanity fails; if $a \neq 0$, its magnitude changes by factors of 3 between all-equal and NAE block inputs, contradicting $|q_T| = 1$. Partial twists and raw orientation-dependent T are outside the theorem.

Exact additive reformulation. Let $\mathcal{T} = \{0, e_0, \ldots, e_{13}\}$ and $\Delta = E_1E_3(E_0 + E_2) = -P_6$. For real multilinear R, $F = P + R$ is Boolean, has degree at most 5, and agrees with P on $\mathcal{T}$ if and only if

$$R = \Delta + L \ (\deg L \le 5), \quad R(R + 2P - 1) = 0 \text{ on } \{0,1\}^{14}, \quad R|_{\mathcal{T}} = 0.$$

Every admissible F occurs exactly once, with $R = F - P$.

Exact XOR reformulation. Every Boolean F occurs uniquely as

$$J = P \oplus F, \qquad F = P + (1 - 2P)J.$$

The desired functions correspond exactly to Boolean masks J with $J|_{\mathcal{T}} = 0$ and $\deg_{\mathbb{R}}(P + (1 - 2P)J) \le 5$. Necessarily, but not sufficiently, $\mathrm{ANF}_{\ge 6}(J) = \mathrm{ANF}_{\ge 6}(P)$. The additive and XOR systems are lossless changes of coordinates, not restricted search families.

The simplest mask fails at degree 10. For $J_0 = g_1g_3(g_0 \oplus g_2)$,

$$\deg_{\mathbb{R}} J_0 = 8, \quad (J_0)_8 = -2E_0E_1E_2E_3,$$

whereas exact integer Möbius inversion gives

$$\deg(P \oplus J_0) = 10, \quad (P \oplus J_0)_{10} = 4E_0E_1E_2E_3x_{12}x_{13}.$$

The top defect has exactly $3^4 = 81$ monomials, all coefficient $+4$.

Status and review focus. No degree-five, sensitivity-fourteen construction, improved recursive exponent, or priority claim is made. The accompanying standard-library verifier reconstructs $K, A, P, J_0, P \oplus J_0$ and checks their exact integer Möbius coefficients. Specialist review should focus on both signature lemmas, the selector averaging, gate-factor and parity scopes, both exact systems, the degree-ten calculation, and the literature comparison.